Standard Chemotherapy Subset Analysis

Comparison of Transition Matrices and Petri Net Simulations
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1. Background

The current combined transition matrix is based on 240 paired samples from multiple GEO datasets. However, these datasets mix different treatment regimens (chemotherapy, targeted therapy, immunotherapy, endocrine therapy). This analysis isolates samples that received standard neoadjuvant chemotherapy (primarily anthracycline + taxane) to examine whether the immune state transition dynamics differ under a more homogeneous treatment context.

2. Data Source

GSE32603 (I-SPY1 Trial): Most patients received anthracycline-based chemotherapy followed by a taxane. Treatment codes F1 and FS were used as proxies for standard chemotherapy.

- Standard chemo paired samples extracted: ~43 patients
- Valid pairs used for transition matrix: 83 (multiple samples per patient in some cases)

3. Transition Probability Comparison

Table 1. Transition Probabilities — Standard Chemotherapy vs Full Cohort

From \ To	Low (Std)	Mid (Std)	High (Std)	Full Cohort High
Low	0.393	0.607	0.000	0.122
Mid	0.500	0.214	0.286	0.286
High	0.296	0.444	0.259	0.420

Key observations:

- Low → Mid is substantially higher under standard chemo (0.607 vs ~0.32 in full cohort)
- Low → High is zero under standard chemo, vs 0.122 in the full cohort
- High state is less stable under standard chemo (0.259 vs 0.420)

4. Petri Net Simulation Comparison

500 simulations were run using the same I-SPY2 pCR probabilities for Step 1, but different immune transition matrices for Step 2.

Table 2. Post-treatment Immune State Distribution (500 simulations)

Next Immune State	Standard Chemo	Full Cohort (v7)
Low	206	231
Mid	**210**	145
High	**84**	124

Interpretation:

- Under standard chemotherapy, post-treatment states concentrate more in Mid
- High immune state is less frequently reached or maintained

- This is consistent with the transition matrix: Low tends to move to Mid, and High is less stable

5. Limitations

- Small sample size for the standard chemo subset (~43 patients)
- Treatment codes (F1, FS) are proxies; exact regimens are not fully documented in the public data
- Platform and batch differences across datasets remain
- No direct comparison of pCR rates by regimen within the paired GEO data

6. Conclusion

Restricting the analysis to standard chemotherapy produces a visibly different immune transition pattern compared with the mixed full cohort. The standard chemo subset shows stronger movement from Low toward Mid and weaker retention of High states. These differences suggest that treatment regimen is a meaningful modifier of state dynamics, and that future models should either stratify by regimen or explicitly include regimen as a factor in the Petri net structure.

Files generated:

- gse32603_standard_chemo_paired.csv
- two_step_simulation_std_chemo.csv